



Date: 19-09-2024

DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
(An Institution of National Importance under Ministry of Education, Govt. of India)
MID SEMESTER EXAMINATIONS-I
B. Tech I Semester
MAIC-101: Differential Calculus and Differential Equations
Time: 09:30 AM to 10:20AM
Max. Marks: 20

N.B.: Answer ALL questions.
Answers to all parts of each question should be in one place.

Q1	Let $P(x) = a_0 + a_1x + a_2x^2$. Choose (find) a_i such that $P(1) = 1, P(2) = 2$ and $P(3) = 3$. Is this choice unique? [5M]
Q2	Check $U = \{(1,1,0,0), (0,1,1,0), (0,0,1,1), (1,0,0,1)\}$ is linearly independent or linearly dependent over field $\mathbb{R}$, with proper explanation. [5M]
Q3	Consider the quadratic form $Q = x^2 + y^2 + z^2 + 4xy + 4yz + 4zx$ in three variables. [10M] ☒ a) Express $Q = X^TAX$ ☐ b) What type of matrix A is ☒ c) Find the Eigen values of A ☐ d) Find the Orthonormal matrix that converts Q into its canonical form ☒ e) Express the new co-ordinates in the canonical form in terms of the old co-ordinates ☒ f) What is the nature of the quadratic form ☒ g) Evaluate A^{-1} using Cayley-Hamilton theorem